```
> restart;#Microwave
> # http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/h21.html
> #10.9Ghz is 3.59eV on the microwave thus we can deduce the
  Planck's constatnt for miccrowave;
> E:=hmw*nu;
```

$$E := hmw\,\nu \tag{1}$$

```
> hmw:=3.59/(10.9*10^9);
```

$$hmw := 3.293577982 \times 10^{-10} \tag{2}$$

```
> 160.23*10^9 ;#Penzias and Wilson originally detected the Cosmic
  Microwave Background 160.23GHz with an energy of ~7.1eV ;
```

$$1.602300000 \times 10^{11} \tag{3}$$

```
> c := 299792458;
  160.23*10^9;
  evalf(c/(1.9*10^11));evalf(c/(10.9*10^9));
```

$$c := 299792458$$

$$1.602300000 \times 10^{11}$$

$$0.001577855042$$

$$0.02750389523 \tag{4}$$

```
> hmw:=0.188977/(1.057*10^9);13.60569292/4;3.5904-3.401423;
```

$$hmw := 1.787861873 \times 10^{-10}$$

$$3.401423230$$

$$0.188977 \tag{5}$$

```
> h := 6.6260701500*10^(-34);rb:=5.239*10^(-10);
  c := 299792458;va := 343;b:=evalf(c/va)^2;
  T := 273.15 ;ec := 1.6021766340*10^(-19);Tq:= 1.765*10^(-19);
  k := 1.3806490000*10^(-23);
```

$$h := 6.626070150 \times 10^{-34}$$

$$rb := 5.239000000 \times 10^{-10}$$

$$c := 299792458$$

$$va := 343$$

$$b := 7.639292971 \times 10^{11}$$

$$T := 273.15$$

$$ec := 1.602176634 \times 10^{-19}$$

$$Tq := 1.765000000 \times 10^{-19}$$

$$k := 1.380649000 \times 10^{-23} \tag{6}$$

```
> 0.19/c/(-364.64^(-1)+345.35^(-1))*10^(-9);he:=h/ec;solve(0.19/c/
  (-x^(-1)+345.35^(-1))*10^(-9)=he);
```

$$4.137370747 \times 10^{-15}$$

$$he := 4.135667697 \times 10^{-15}$$

$$364.6483874 \tag{7}$$

```
> http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/Lamb2.html#c1
> 160.23*10^9*hmw;#The energy of CMB
```

$$28.64691079 \tag{8}$$

```
> 1239.84/28.64691079;
```

$$43.28005938 \tag{9}$$

```
> 2.395*10^9*hmw;
```

$$0.4281929186 \tag{10}$$

```
> 3/2*k*27777/ec
```

$$3.590454990 \tag{11}$$

```
> exp(3.59/(k*T*b))
```

$$6.113721285 \times 10^{541179212} \tag{12}$$

```
> (4*r^2*8*π^2 *ν^2)/c^2 ;
```

$$\frac{8\, r^2\, \pi^2\, \nu^2}{22468879468420441} \tag{13}$$